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# Soil loss in an olive grove in Central Spain under cover crops and tillage treatments, and farmer perceptions

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**Abstract**
*Purpose* Soil erosion is a threat for the sustainability of the grove production all over the world, and olive groves are where the highest erosion is achieved. This study aimed to (1) evaluate soil loss from water erosion in microplots (1 $m^2$) under natural rainfall events in an olive grove managed with tillage and three different cover crops; (2) determine the main climatic factors on soil loss and selective transportation; and (3) survey the willingness of local farmers to adopt cover crops.
*Materials and methods* Over four hydrological years (2010 to 2014), tillage and three cover crop treatments have been assessed in a sloping olive grove in the Mediterranean area, in a semiarid climate and gypsiferous soil. The treatments were tillage (once a year), two annual covers (barley and legumes) and a permanent cover (*Brachypodium distachyon*). Soil loss was collected after each rainfall event, dried and weighted. Rainfall data was recorded and soil cover (%) was determined. Organic carbon (% OC) and texture were measured for sediments of tillage treatment to study their selective transportation. Local farmers were polled regarding their knowledge of the biophysical factors that affect soil conservation, their management practices and their willingness to adopt sustainable land management practices as cover crops.
*Results and discussion* The tillage treatment exhibited the highest soil loss (6.8 t $ha^{-1}$ $year^{-1}$). Cover crops significantly reduced soil loss: 40 % for legumes, 60 % for barley and permanent cover led to an 80 % decrease (1.4 t $ha^{-1}$ $year^{-1}$). The soil loss depends mainly on the soil cover and kinetic energy of rainfall events. Vegetation cover higher than 40 % in autumn and spring was essential to limiting soil loss, even under heavy events. Mobilised sediments were enriched 2.4 times in OC and 1.6 in the clay fraction. In spite of the well-known benefits of cover crops in soil conservation, local farmers do not use them. A lack of environmental education and awareness has been detected.
*Conclusions* Tillage in olive groves produced unsustainable rates of soil loss, while permanent cover demonstrated high efficiency. Gypsiferous soils are vulnerable to water erosion processes if they are not protected by vegetation. Farmers should adopt tailored soil cover practices during spring and autumn, when the soil loss is higher. A few farmers use sustainable soil management techniques, and therefore cover crops in olive groves are rare in Central Spain because farmers are reluctant to change.

**Keywords** Cover crops · Erosion · *Olea europaea* · SLM practices · Social approach

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## 1 Introduction

Soil erosion is a worldwide concern (Gessesse et al. 2015; Karamesouti et al. 2015; Ligonja and Shrestha 2015; Ochoa

et al. 2016) and one of the most important land degradation driving processes in Mediterranean areas (EEA 2015; Panagos et al. 2015). Although forest lands used to have sustainable soil erosion rates, the impact of agriculture results in high and non-tolerable soil losses (Verheijen et al. 2009; Zema et al. 2016). Vineyards, olive, citrus, persimmon and apricot plantations are seeing as the source of sediments due to the intense tilling, the use of herbicides and the lack of vegetation cover (Keesstra et al. 2016; Prosdocimi et al. 2016 Rodrigo-Comino et al. 2016). Soil erosion is even worse in soils with low concentrations of organic matter and weak structure (Panagos et al. 2014); these erodible soils are common in Spanish olive groves (Taguas and Gómez 2015).

Spain is the major olive producer worldwide with a total area of $2.5 \times 10^6$ ha, primarily in Andalusia (Southern Spain), from a total of $10.3 \times 10^6$ ha in the world (FAOSTAT 2015). There are more than 28,000 ha of olive groves in the Madrid region (MAGRAMA 2013). Although this area is not substantial in relation to the total surface area of Spanish olive groves, it is considered important in the southern section of Madrid. Olives occupy 20 % of the agricultural area of the region, being second in agriculture extension after winter cereals (IECM 2013).

As with other Spanish traditional olive groves, the Madrid olive trees are grown in sloping areas on poor and shallow soils (Gómez et al. 2009b). Ninety-eight percent of the olive groves in the region are rainfed, 50 % are planted at low densities (less than 100 trees $ha^{-1}$) and 80 % are usually tilled to reduce weed competition for water (MAGRAMA 2013). In this semiarid area, water availability is a major flaw, production limitations have been found for vineyards in this region (Ruiz-Colmenero et al. 2011) and the effects of cover crops in olive grove production are considered in an on-going project (Sastre et al. in preparation). Conventional practices involving tillage result in a large area of bare soil that is prone to erosion processes.

Since the 1990s, several studies have demonstrated the efficiency of cover crops in comparison with conventional tillage in olive groves to reduce soil erosion and improve water storage and physical properties. Table 1 summarises soil loss results of several studies under different management: conventional tillage and cover crops. Different studies stated that high-energy storms determined the annual soil loss (e.g. Jacinthe et al. 2004; Bienes and Marques 2008; Martinez-Mena et al. 2012; Taguas et al. 2013). Soil loss also depends on soil cover; for example, Marques et al. (2008) found that when cover exceeds 60 %, rainfall erosivity declined drastically. When linked to the sediments generated by water erosion, certain nutrients and types of organic matter are also lost according to the study conditions (e.g. Bienes et al. 2010; Martínez-Mena et al. 2012).

In Andalusia, cover crops in olive groves appear to be accepted by farmers (Franco and Calatrava-Leyva 2006; Calatrava-Leyva et al. 2007). In spite of the benefits of cover crops in soil as have been verified in the abovementioned studies and also by some studies in vineyards (Ruiz-Colmenero et al. 2013), this is not an extended sustainable land management (SLM) practice among the land users of Central Spain due to the risk of competition for water and nutrient resources (Celette et al. 2008; Gómez et al. 2009a; Alcántara et al. 2011). In the few cases that cover crops have been adopted, the decision is related to economic and labour savings in the short term instead of environmental concerns (Calatrava-Leyva et al. 2007). This fact reveals the need to strengthen environmental education and SLM capacitation in the region. On this matter, the agrarian extension services of the area play an important role in the dissemination of benefits and constraints about cover crops.

Cover crops were tested in a trial performed in a sloping rainfed olive grove under semiarid conditions on a degraded *Haplic Gypsisol*. The study plot is located in the shire with most olive groves of Madrid Region (IECM 2013), under representative edaphoclimatic conditions. Due to the extreme climatic conditions that occur periodically in the area, the adoption of cover crops can be controversial; therefore, possibilities of adoption are examined in the present study. It is hypothesised that cover crops can reduce soil loss mainly under high-intensity rainfall events and that the most effective cover crop would be the permanent one.

## 2 Material and methods

### 2.1 Study area

This study was performed in an experimental olive grove (*Olea europaea* L.) located in Central Spain, in southern Madrid (40° 4′ 21″ N, 3° 31′ 11″ W). The average elevation is 540 m.a.s.l., and the slope ranges from 9 to 12 %. The soil is classified as a *Haplic Gypsisol* (IUSS Working Group WRB 2014), with a xeric moisture regime. This soil has a high silt concentration (29 %), and it is low in organic carbon (0.62 %) and clay (12 %). The climate is semiarid Mediterranean, with long hot summers (quite often above 35 °C in July and August) and cold winters (2.7 °C on average in December). The average annual temperature is 13.6 °C. The annual precipitation is approximately 390 mm with high inter and intra-annual variability (Elías-Castillo and Gimenez Ortiz 1965; Urbano 1992). During the last 17 years, the total precipitation has strongly decreased (40 %) in comparison with a previous period (1933–1969), and at the same time, rainfall intensity increased (60 %) in the study area (García-Díaz et al. 2015).

The olive plantation was established in 2004 with trees in 6 × 7 m spacings, with an area of approximately 3 ha. The cultivar is Cornicabra, the most common in Central Spain (Rallo et al. 2005). The average tree size measured with a

**Table 1** List of publications regarding erosion assessments carried out in olive groves

| Authors (years) | Country | Cover crop type | Soil type | Slope (%) | Soil texture | Total rainfall (mm year$^{-1}$) | Soil loss (t ha$^{-1}$ year$^{-1}$) | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | CT | CC |
| Fleskens and Stroosnijder (2007) | Portugal | Spontaneous | nd | 18 | Sandy loam | 600–800 | 3.3 | 0.1 |
| | Italy | Spontaneous | | 11–38 | Clay loam | 676–866 | 0.5–4.45 | 0.82–1.94 |
| | Italy | Permanent | nd | 32 | Candy clay | 676–866 | 0.2 | ≈0 |
| | Spain | Spontaneous | Inceptisols and Aridisols | nd | nd | 500–600 | 44 | 10 |
| Duran-Zuazo and Rodriguez-Pleguezuelo (2008) | Spain | Barley | Typic Xerorthent | 30 | Loam | 383 | 17.3 | 5.0 |
| Pastor (2008) | Spain | Cereal | nd | moderate | nd | Simulated rainfall | 10 | ≈0 |
| Gómez et al. (2009a) | Spain | *Lolium* sp. | Petrocalcic Palexeralf | 11 | Sandy loam | 576 | 19.4 | 0.4 |
| Gómez et al. (2009b) | Spain | Barley | Typic Haploxerept | 13.4 | Heavy clay soil | 577 | 2.9 | 0.8 |
| Gómez et al. (2011) | Spain | Seasonal grass cover | nd | 4 | Clay loam | 550 | 1.3 | 0.04 |
| Espejo-Perez et al. (2013) | Spain | Spontaneous | Calcic Haploxerept | 17 | Clay loam | 689[a] | 16.8[b] | 11.8[b] |
| | | Spontaneous | Calcic Haploxerept | 18 | Loam | 834[a] | 31.6[b] | 9[b] |
| | | Spontaneous | Ruptic Xerorthent | 18 | Silt loam | 1016[a] | 42.3[b] | 11.6[b] |
| | | Spontaneous | Typic Calcixerept | 7 | Silty clay loam | 673[a] | 36.6[b] | 13.5[b] |
| | | Spontaneous | Typic Haploxerert | 18 | Clay loam | 628[a] | 99.3[b] | 32.9[b] |
| | | Spontaneous | Typic Xerochrept | 6 | Loam | 496[a] | 75.1[b] | 20.5[b] |
| | | Spontaneous | Typic Xeropsamment | 8 | Loam | 770[a] | 89.1[b] | 61.6[b] |
| | | Spontaneous | Typic Xeropsamment | 9 | Clay loam | 770[a] | 86.9[b] | 21.3[b] |
| Kairis et al. (2013) | Greece | Spontaneous | Calcaric Cambisol | 17 | Clay loam | 512 | 0.1–0.4 | 0.01–0.05 |
| Marquez-Garcia et al. (2013) | Spain | Spontaneous | Calcic Haplowerept | 15.6 | Clay loam | 500 | 1.3 | 0.6 |
| | Spain | Spontaneous | Ruptic-Lithict Xerorthent | 21.6 | Silty loam | 565 | 2.1 | 0.2 |
| | Spain | Sown | Calci Haploxerept | 18.6 | Clay loam | 416 | 8.2 | 1.4 |
| | Spain | Spontaneous | Typic Calcixerept | 6.2 | Loam | 501 | 2.5 | 0.3 |
| | Spain | Spontaneous | Typic Haploxerert | 8.7 | Clay loam | 535 | 3.7 | 0.3 |
| Palese et al. (2015) | Italy | Spontaneous grasses | Haplic Calcisol | 0–16 | Sandy loam | Simulated rainfall | 0.005[b] | ≈0[b] |

*nd* not available data, *CT* conventional tillage, *CC* cover crops

[a] Cumulative rainfall (mm)

[b] Units are tons per hectare

surveying rod in 10 olive trees was 1.9 m high and 2.8 m in crown diameter. The crown projection was calculated using the ellipse model and measuring two diameters: maximum projection axis and its perpendicular axis; thus, the soil surface covered by the tree's crown is close to 8 %. Before this study, the olive grove was managed by tillage (using a chisel up to 0.30 m deep, in spring and autumn) to decrease weed competition, and it was grown in rainfed conditions. The study started in autumn (November) 2010 and lasted until October 2014.

There is an automatic weather station in the olive grove to record the temperature, rainfall amount, relative humidity, solar radiation, wind speed and direction and atmospheric pressure every 10 min. For each rainfall event, the following measures were determined: total rainfall amount ($P$), maximum 15-min rainfall intensity ($I_{15}$), maximum 30-min rainfall intensity ($I_{30}$), maximum hourly rainfall intensity ($I$), storm kinetic energy (KE) and rainfall erosivity ($R$). The KE and $R$ were calculated as follows (Wischmeier and Smith 1978):

$$\mathrm{KE}\left[\mathrm{MJ\ ha^{-1}}\right] = 0.119 + 0.0873 \times \mathrm{Log}_{10} I \quad (1)$$

$$R\left[\mathrm{MJ\,mm\,ha^{-1}\,h^{-1}}\right] = \mathrm{KE} \times I_{30} \quad (2)$$

The classification of rainfall events based on KE data from Jacinthe et al. (2004) was used. Thus, the rainfall events were divided into (i) class 1, in which KE < 2 MJ ha$^{-1}$ are low amount and low intensity; (ii) class 2, in which

2 < KE < 4 MJ ha$^{-1}$ are moderate-to-high amount and moderate intensity; and (iii) class 3, for which KE > 4 MJ ha$^{-1}$ are high amount and high-intensity storms. The seasons were considered as follows: spring, from 21 March to 21 June; summer, from 21 June to 21 September; autumn, from 21 September to 21 December; and winter, from 21 December to 21 March.

### 2.2 Olive grove management

In November 2010, the whole area was ploughed with a chisel to a depth of approximately 0.30 m to follow the traditional management practice in the area. The tillage was performed in 6-m-wide inter-rows, at the centre of the rows between the olive trees (Fig. 1). Each treatment was performed in three consecutive inter-rows. The treatments consisted of the following: two annual covers, (1) *Hordeum vulgare* L., barley (HOR) that was seeded each autumn at 70 kg ha$^{-1}$, and (2) legume (LEG), sainfoin (*Onobrychis viciifolia* Scop.) that was seeded each year between 2010 and 2013 (42 kg ha$^{-1}$). However, because of a progressive decline in the soil cover in dry seasons, the sainfoin was replaced in the last year by another leguminous plant, namely bitter vetch (*Vicia ervilia* Willd.), at 75 kg ha$^{-1}$. The seeding took place at the same time as that of HOR. (3) *Brachypodium distachyon* (BRA) treatment consists of permanent grass cover with purple false brome (*B. distachyon* L. P. Beauv.) at 40 kg ha$^{-1}$. *B. distachyon* is a grass species native to southern Europe, northern Africa and southwestern Asia, which sprouts in early spring, being already mature and dry by late June, at which point it is self-sowed and sprouted spontaneously again the following autumn. (4) Control (CON) consisted of one pass per year with a chisel at 0.15–0.20 m deep in mid-November, to control weeds. Cover crops were seeded (the annual cover crops were seeded once a year and purple false brome was seeded once during the study period) with a grain drill that had harrow tines and without fertilisation. To maintain bare soil along the olive tree line, a contact herbicide (glyphosate) was applied to the soil only in the rows of olive trees twice per year. As a result, 1-m width along the line was bare to facilitate olive harvesting. All the treatments were mowed once in the spring (during the first fortnight of May), except in 2013, when the vegetation was mowed twice (the second cut was by the end of May) due to a high vegetation growth resulting from the abundant rains of that spring. Plant debris was left spread on the surface.

Cover crop and soil variables were studied in microplots established at the centre of the olive rows. Microplots are metal sheet-bounded plots that cover 1 m$^2$ (0.5-m wide and 2-m long; Fig. 1). Three microplots were set up per treatment (4 treatments × 3 plots = 12 in total).

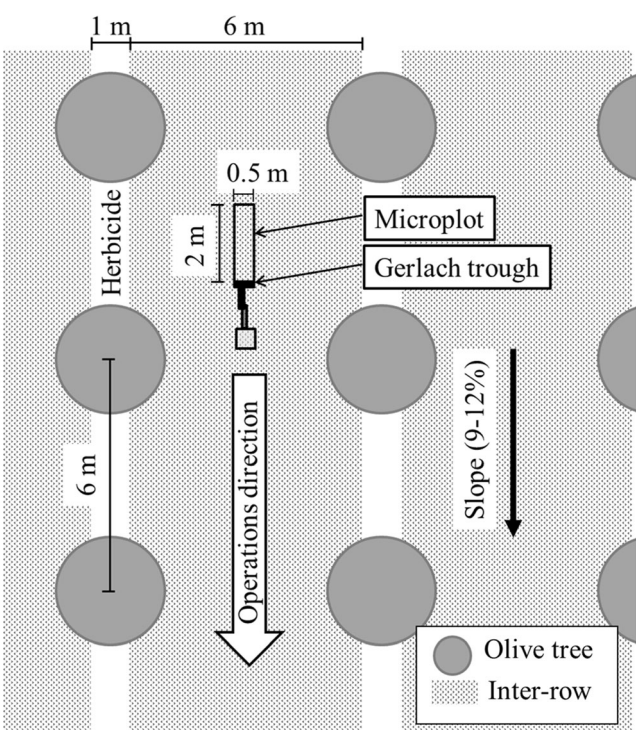


**Fig. 1** Layout of olive trees positions, primary operations and microplot locations in the olive grove

### 2.3 Soil cover

The soil cover in each microplot was determined bi-monthly with 25-cm × 25-cm quadrats. Cover was estimated by six trained observers, and the average value of the six observations was obtained. The results of this method are similar to those obtained by García-Estríngana et al. (2005) employing digital image analysis of coloured orthogonal photos of vegetation cover and are cost and time effective.

### 2.4 Sediment analysis

A Gerlach trough (Gerlach 1967) was placed at the base of each microplot to collect sediment yield. Sediments (from January 2011 to September 2014) were collected from the Gerlach trough after each rainfall event. The sediments were oven-dried at 105 °C and weighed.

To study the particle size-selective transport, we assessed the texture composition and organic carbon (OC) from 4 points along the $Ap_1$ horizon (approximately 0.12 m deep) in the study area (hereafter known as reference soil) at the beginning of the study. The samples were taken randomly in the olive grove, in locations near the microplots. The texture and OC from ten sediment samples of the CON treatment were also analysed. This treatment was chosen because it provided sufficient sediment to carry the analysis. The selected samples were set after mixing three sediment replicates. Five of these samples were produced by moderate-intensity rainfall events ($I$ < 20 mm h$^{-1}$) according to the classification by Porta

et al. (1999), and the other five came from high-intensity events ($I > 20$ mm h$^{-1}$). The OC analysis was made for eight samples because there was not enough soil in the other two.

The texture was assessed by Robinson pipette method (MAPA 1994), and the OC was determined by using the wet chemical oxidation method (Walkley and Black 1934).

### 2.5 Social approach

One hundred nineteen farmers were polled regarding their knowledge of the biophysical factors that affect soil conservation, their management practices and their willingness to adopt SLM practices as cover crops.

The surveys were performed from May to September 2014. A total of 97 farmers were randomly contacted through three agricultural offices in the study area, the south-eastern of Madrid region. The sample was completed with 22 interviews by following a snowball approach (Faugier and Sargeant 1997). From these farmers, 83 were olive growers.

The questions in the structured questionnaire (see Annex 1: Electronic Supplementary Material) were organised in the following three primary groups: (i) basic characteristics, the study level, age, primary activity and plot size; (ii) knowledge about land degradation; and (iii) information about the farmer's own land management practices.

### 2.6 Statistical procedures

Statistical procedures were performed with the software package SPSS 19 Inc. for Windows. Data were log-transformed before parametric testing. ANOVA analyses were performed for the soil parameters to assess the differences between treatments, seasons, sediments and reference soil. Tukey's test ($p < 0.05$) was used to establish significant differences between groups. When the transformations did not work, a Games-Howell non-parametric test was conducted. Spearman's correlations were employed to calculate the correlation between the measured variables. A linear regression model was established with transformed data to determine which variables best explained soil erosion.

## 3 Results

### 3.1 Rainfall characteristics

From November 2010 to October 2014, the annual average precipitation was 283 ± 40 mm year$^{-1}$. These years were quite similar in terms of monthly temperatures, but not precipitation, which was highly irregular between months and years, as expected in a Mediterranean climate. The driest cropping season was 2011/2012 (237 mm year$^{-1}$), whilst 2012/2013 was the wettest, reaching 328 mm year$^{-1}$. There were great variations in rainfall amounts between spring (49 to 162 mm) and autumn seasons (39 to 155 mm). Summers were quite similar with low rainfall amounts (3 to 22 mm).

Fifty rainfall events were recorded between January 2011 and September 2014. Table 2 shows the primary characteristics of these events. There were significant differences between seasons for $P$, $I_{15}$ and KE. Rainfalls in summer were significantly lower than the rest of seasons. Summer also had the lowest KE and $R$ values, although not statistically different from all other seasons. Seventeen out of the 50 recorded events were in KE class 1 (KE ≤ 2 MJ ha$^{-1}$), 22 of class 2 (2 < KE ≤ 4 MJ ha$^{-1}$) and 11 in class 3 (KE > 4 MJ ha$^{-1}$).

### 3.2 Soil cover

Figure 2 shows the evolution of soil cover percentages for the four treatments. It shows a sawtooth-type evolution from tillage and mowing actions. Once installed, the BRA kept over 40 % of the soil covered despite the mowing. CON always had less than 20 % cover, but the annual crops exhibited different behaviour during different cropping seasons. In 2012/2013, the sainfoin (LEG) cover increased quickly because of a rainy spring, and during the following season, bitter vetch (LEG) worked better than sainfoin during the driest period, reaching 40 % cover during the spring. HOR increased the soil cover during the rainy spring (2012/2013), and for the driest one, the cover was only slightly more than 20 %. After the summer, annual cover crops reduced the soil cover, leaving only stubbles and plant debris.

### 3.3 Soil loss

The annual soil loss caused by water erosion was 6.81, 4.00, 2.74 and 1.36 t ha$^{-1}$ year$^{-1}$ for CON, LEG, HOR and BRA, respectively (Table 3). Soil loss from BRA was significantly lower than loss from CON and LEG, HOR was intermediate. Regarding the soil loss per event, CON had the highest values, followed by the annual cover crops; BRA was the treatment that best controlled the soil loss per rainfall event. LEG had a similar behaviour to that of HOR.

Figure 3 shows the cumulative soil loss for the study period for the different treatments as well as the precipitation per event. The total soil loss over the study period from the CON treatment was 25.8 t ha$^{-1}$;15.5 t ha$^{-1}$ from LEG; 10.6 t ha$^{-1}$ from HOR and 4.7 t ha$^{-1}$ from BRA. Each time a heavy rainfall occurred, soil loss dramatically increased, and it was more pronounced in the CON treatment and in the first year. From the beginning of the study, the soil loss rate of CON was higher than that of the cover crops. The CON was still increasing at the same speed during the second cropping season, and the BRA was separated from annual cover crops. During the 2012/2013 season, when rainfall events were more intense, the behaviour of the CON and BRA treatments did

**Table 2** Average and standard deviation (SD) of the primary characteristics of rainfall events: number of events (*N*), rainfall amount (*P*), maximum 15-min rainfall intensity ($I_{15}$), maximum 30-min rainfall intensity ($I_{30}$), maximum hourly rainfall intensity (*I*), kinetic energy (KE) and rainfall erosivity (*R*)

| | Average | Season | | | |
|---|---|---|---|---|---|
| | | Autumn | Winter | Spring | Summer |
| *N* | 50 | 14 | 16 | 16 | 4 |
| *P* (mm) | 22 ± 15 | 25 ± 17 b | 21 ± 14 b | 24 ± 15 b | 8 ± 4 a |
| $I_{15}$ (mm h$^{-1}$) | 8.8 ± 6.7 | 11.1 ± 10.4 ab | 5.7 ± 2.7 a | 10.0 ± 5.0 b | 8.9 ± 4.3 ab |
| $I_{30}$ (mm h$^{-1}$) | 7.1 ± 4.9 | 8.9 ± 7.8 a | 5.2 ± 2.1 a | 7.0 ± 3.2 a | 8.7 ± 3.4 a |
| *I* (mm h$^{-1}$) | 11.3 ± 9.1 | 13.1 ± 14.4 a | 7.8 ± 3.0 a | 13.3 ± 7.5 a | 11.1 ± 6.5 a |
| KE (MJ ha$^{-1}$) | 3.3 ± 2.5 | 3.4 ± 3.2 ab | 3.2 ± 2.1 ab | 3.9 ± 2.4 b | 1.1 ± 0.6 a |
| *R* (MJ mm ha$^{-1}$ h$^{-1}$) | 16.7 ± 22.5 | 23.6 ± 37.4 a | 11.2 ± 9.7 a | 17.9 ± 15.9 a | 5.1 ± 5.0 a |

Different lowercase letters mean differences between seasons at $p < 0.05$

not change. However, LEG began gaining distance from HOR a year before the change in LEG species, despite having a similar cover percentage. For the last season, the BRA almost fully controlled the soil erosion.

In accounting for the average soil loss per season, the soil loss during the spring was higher than that of the summer (Table 4) because of that season's more intense and erosive rainfall events (Table 2). However, when looking at seasonal differences within treatments, CON was the only one with significant differences between spring and summer. The other treatments did not experience significant differences between seasons. Regarding the differences between treatments within seasons, BRA significantly reduced soil loss with respect to CON and LEG in autumn and winter, whilst in spring, CON had the highest soil loss, significantly different from BRA and HOR.

A correlation among soil losses and the rainfall event characteristics along with the cover percentage in the microplots has been conducted (data not shown). Strong correlations ($p < 0.01$) were found between all the analysed variables and the soil loss. The best correlated variable with soil loss was the cover, with a negative relation ($r = -0.48$). All the others have a positive relation: 0.25 for *P*, 0.27 for $I_{15}$, 0.18 for $I_{30}$, 0.26 for *I*, 0.21 for KE and 0.23 for *R*.

A statistical linear modelling analysis was performed by employing the studied variables and the transformed soil loss data. A model (Eq. 3) with an $R^2$ coefficient of 0.293 ($F = 42.178$, $df = 2$, $p < 0.001$) was obtained. Only two of the seven studied variables had significance in the model: cover was the most important predictor, with a weight of 0.76, and KE at 0.24. The rest of climatic variables did not help to explain the dependent variable because of their strong correlation with KE.

$$\text{Soil loss} = e^{2.865-0.023\times\text{Coverage}+0.141\times KE} - 1 \quad (3)$$

Because of the importance of the two variables in explaining soil loss, a more detailed analysis was carried out.

As shown in Fig. 4, a good adjustment with a logarithmic equation between soil loss reduction and ten categories of cover was found ($R^2 = 0.81$). When more soil is covered, less soil is lost. This relationship is stronger with low percentages of covered soil, in the first half of the curve.

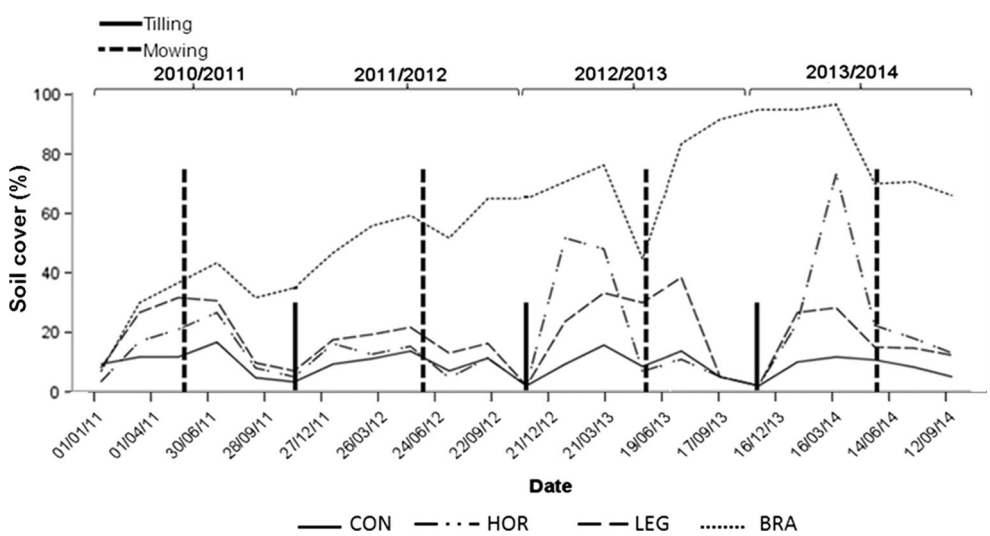


**Fig. 2** Soil cover (%) per treatment (*CON* control, *HOR* barley, *LEG* legume and *BRA B. distachyon*) over the study time. Tilling and mowing operations are indicated

**Table 3** Average and standard deviation (SD) of soil loss for the treatments control (CON), barley (HOR), legume (LEG) and *B. distachyon* (BRA); the reduction percentage of soil loss compared with the control treatment; and the average and SD of soil loss per event

| Treatment | *N* | Annual soil loss (t ha$^{-1}$ year$^{-1}$) | Reduction percentage of soil loss respect to CON (%) | *N* | Soil loss per event (g m$^{-2}$) |
|---|---|---|---|---|---|
| CON | 12 | 6.81 ± 6.49 b | – | 150 | 51.1 ± 82.5 c |
| HOR | 12 | 2.74 ± 2.53 ab | 59.77 | 150 | 21.1 ± 44.1 b |
| LEG | 12 | 4.00 ± 2.58 b | 41.26 | 150 | 31.1 ± 42.0 b |
| BRA | 12 | 1.36 ± 1.58 a | 80.03 | 150 | 10.0 ± 24.0 a |

*N* is the number of cases (4 cropping seasons × 3 repetitions = 12 and 50 rainfall events × 3 repetitions = 150). Different letters mean significant differences between treatments at $p < 0.05$

The KE was the other relevant variable in relation to soil loss in the regression model. A linear relationship was found between the cumulative soil loss and the cumulative KE per treatment (Fig. 5). A high correlation was found in every case ($R^2 > 0.91$) with a higher slope for CON treatment, followed by LEG and HOR, and BRA had the smallest one, meaning that it had the lowest soil loss rate in function of KE.

Only 11 of 50 recorded rainfall events (22 %) were high-energy events (KE > 4 MJ ha$^{-1}$), producing 30–42 % of the total mobilised sediments, depending on the treatment (Table 5). The average soil loss values within treatments with respect to KE class 1 and class 2 events were quite similar, whilst class 3 had the highest values. The only treatments for which there were significant differences ($p < 0.05$) between class 3 and the other classes were CON and LEG.

With regards to soil loss between treatments, the BRA was the best treatment, reducing the soil loss in each class of KE (Table 5) and controlling almost all the soil loss. For classes 1 and 2, soil loss was significantly lower ($p < 0.05$) for BRA regarding LEG and CON treatments. BRA also was the best treatment controlling soil loss under class 3 events, significantly better than LEG and CON. In this case, there were also significant differences ($p < 0.05$) between HOR and CON.

In Table 6, the soil loss is shown within three cover groups that are independent of the treatments, namely low (0–20 %), medium (20–40 %) and moderate-high (>40 %) for the three classes of rainfall KE. High-energy storms with scarce cover yield the highest rates of soil loss (76.7 g m$^{-2}$ on average). Classes 1 and 2 produced significantly higher rates of soil loss under low cover than when the cover was above 40 %. For class 3 rainfall events, low cover produced significantly more soil loss than medium and moderate-high cover groups. With medium and moderate-high cover, the rainfall events do not seem to generate an increase in soil loss with a higher KE, although there is a trend in the results.

### 3.4 OC concentration and texture of sediments

The relationship between the OC and the texture of sediments and the primary characteristics of rainfall events were studied by using their degree of correlation (Table 7). Significant and positive relations were found for OC and both silt and clay + silt fractions ($p < 0.05$); therefore, the fine fractions and OC were linked. Sand had a negative correlation with $I_{15}$, $I_{30}$ and $I$, whilst fine fractions (silt, clay or both) had a positive correlation with several rainfall properties, primarily with the rainfall intensities. As expected, the OC mobilisation was also related to rainfall intensity, in this case with $I$ and $I_{30}$ ($p < 0.10$) (Table 7).

In Table 8, the results for the textural fractions and the OC of the sediments were grouped according to the intensity of the event that produced the soil loss. Moderate events have intensities of less than 20 mm h$^{-1}$, and high-intensity events were between 20 and 50 mm h$^{-1}$. There is a 40 % increase in the clay concentration in moderate-intensity sediments relative to the reference, and there is a statistically significant increase of 80 % in sediments from high-intensity events. Although there were no significant differences, a decreasing

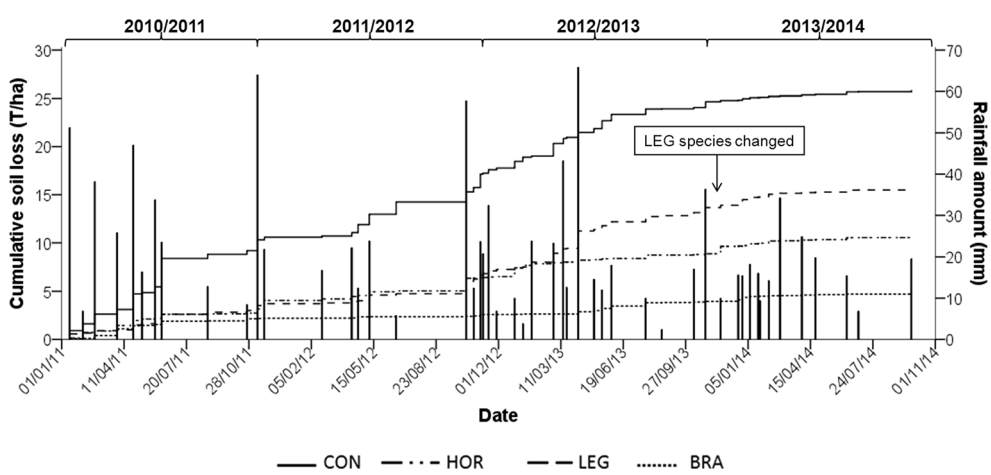


**Fig. 3** Cumulative soil loss (t ha$^{-1}$) per treatment (*CON* control, *HOR* barley, *LEG* legume and *BRA B. distachyon*) and the rainfall amount (mm) registered per event in the columns

**Table 4** Average and standard deviation (SD) of soil loss per event (g $m^{-2}$) during the different seasons for the treatments control (CON), barley (HOR), legume (LEG) and *B. distachyon* (BRA)

| Treatment | Seasons | | | |
|---|---|---|---|---|
| | Autumn (14) | Winter (16) | Spring (16) | Summer (4) |
| CON | 45.1 ± 42.7 ab B | 39.8 ± 41.2 ab B | 75.7 ± 73.0 b B | 25.7 ± 28.0 a A |
| HOR | 27.2 ± 41.4 a AB | 19.3 ± 23.5 a AB | 20.3 ± 17.4 a A | 10.4 ± 17.1 a A |
| LEG | 33.8 ± 30.4 a B | 26.5 ± 24.8 a B | 35.8 ± 43.3 a AB | 21.3 ± 23.1 a A |
| BRA | 5.5 ± 6.9 a A | 5.7 ± 8.6 a A | 16.4 ± 18.2 a A | 10.1 ± 16.2 a A |
| Mean | 27.9 ± 35.7 ab | 22.8 ± 29.2 ab | 37.1 ± 49.3 b | 16.9 ± 20.6 a |

Different lowercase letters in each row mean significant differences between seasons. Different uppercase letters in each column mean differences between treatments at $p < 0.05$. Number of rainfall events in brackets

trend in the sand fraction under moderate and high rainfall events appeared.

The OC was 2 and 2.5 times higher in sediments (moderate and high-intensity groups, respectively) than in the original soil. The difference was statistically significant between the reference and high-intensity sediment groups.

### 3.5 Farmer perceptions

The number of farmers who participated in the survey represents 10 % of the farmers in this south-eastern county of Madrid, according to the last census with district-level reporting (INE 2002). Table 9 shows the different results of the survey. The age of farmers working in that region may be considered high, at an average of 54 years old. The participants highlighted that agriculture is not their primary source of income. They often declare that they have not received any education or training about soils, and most of them continue following family traditions when managing their crops. Nearly all of them (95 %) receive subsidies to support their farming activities. However, the number of subsidies related to soil conservation or ecological practices is below 17 %. Soil management is primarily based on tradition, and the farmers literally practice 'what has always been done'. After tradition, the reason for selecting a particular management practice is economic profit, and the third reason is based on product quality followed by environmental reasons. Contrary to other studies, only 7 % of farmers cite subsidies as an important push factor to adopt a particular farming decision.

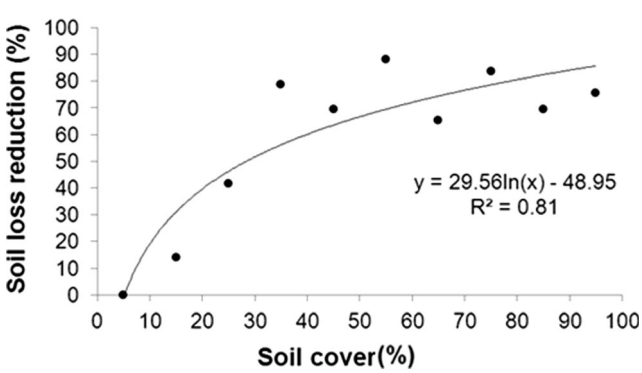


**Fig. 4** Soil loss reduction compared with covered soil

When considering soil conservation problems in order of importance, the farmers were concerned about steep slopes, waterlogging, erosion, low water retention, low fertility, compaction, soil organic matter depletion, pollution, salinization, alkalinity and acidity. They believe that they can help to solve soil problems (79 %), especially by increasing soil organic carbon (SOC), fertilisers and even pesticides, but not so often by improving their ploughing method (Table 9). They declared that they would need better training and improved technology. Government aid ranks in third place among their requirements for improving soil management.

The sources of information for farmer professional activities are primarily regional extension services and farmer neighbours and acquaintances. Internet is providing useful information for farmers, and it is more important than commercial agricultural suppliers or farmer associations. A few farmers (<4 %) use specialised publications, newspapers or university professionals if they are looking for advice.

Only 5 out of the 83 olive growers who took the survey used cover crops on their land and they do so for economic reasons, not due to environmental concerns. Two farmers describe how they let spontaneous vegetation grow to save labour and fuel costs; they use herbicides in the spring to avoid water competition, but they do not consider this practice to be cover cropping. Moreover, the growers mention a social disapproval towards those who 'do not work the soil'; they frequently mention that letting vegetation grow in the olive groves 'is for lazy farmers'.

## 4 Discussion

The effect of cover crops assessed in this study showed different efficiency in controlling soil loss, but all of them achieved better results than the CON supporting our hypothesis. On average, the reduction in annual soil loss in comparison with the CON was 40, 60 and 80 % for LEG, HOR and

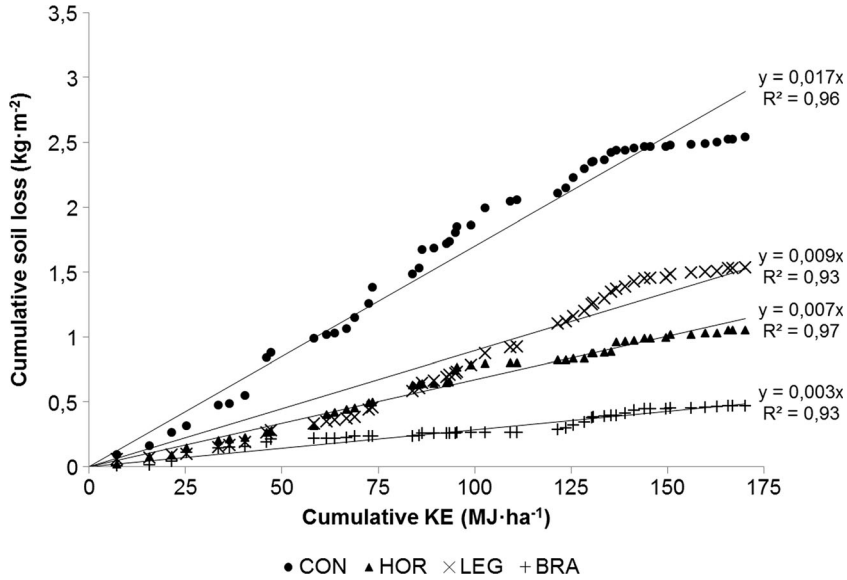


**Fig. 5** Linear regression between cumulative KE and cumulative soil loss for the different treatments (*CON* control, *HOR* barley, *LEG* legume and *BRA B. distachyon*)

BRA, respectively (Table 3), similar to the results of Espejo-Perez et al. (2013) who found a soil loss reduction of 76 % on average.

This finding is consistent with reductions in soil loss from plant cover as reported in other olive-growing regions in Spain, which were between 56 and 91 % (Ordóñez-Fernández et al. 2007; Marquez-Garcia et al. 2013). Gómez et al. (2009b) also found that once cover crops are established, they reduce runoff and sediment concentrations more efficiently than conventional tillage.

Our results for the soil loss of cover crops (1.36, 2.74 and 4.00 t ha$^{-1}$ year$^{-1}$ for BRA, HOR and LEG, respectively) are close to those measured by Taguas et al. (2013) in an olive microcatchment under no-tillage treatment with a spontaneous grass cover in a vertic soil (1.8 ± 2.3 t ha$^{-1}$ year$^{-1}$) and above those found by Gómez et al. (2009b), who recorded 0.4 t ha$^{-1}$ year$^{-1}$ on average in an olive orchard on a sandy loam soil.

The actual soil erosion rates in Europe ranged from 3 to 40 t ha$^{-1}$ year$^{-1}$ (Verheijen et al. 2009). Gómez et al. (2009b) measured 19.4 t ha$^{-1}$ year$^{-1}$ under conventional tillage, which was almost threefold higher than the rates found by our team (6.81 t ha$^{-1}$ year$^{-1}$), because of differences in the high rainfall amount (576 mm on average), soil type and scale of measurements.

Tolerable soil loss depends on driving factors, namely, weathering (e.g. the soil type, climate, and land use) and dust deposition (Verheijen et al. 2009), with a proposed European

**Table 5** Cumulative soil loss for the treatments control (CON), barley (HOR), legume (LEG) and *B. distachyon* (BRA); percentage of soil loss relative to the total soil loss within the treatment; and average and standard deviation (SD) of soil loss per event for each management and rainfall kinetic energy (class 1: KE ≤ 2 MJ ha$^{-1}$, class 2: 2 < KE ≤ 4 MJ ha$^{-1}$ or class 3: KE > 4 MJ ha$^{-1}$)

| Treatment | Rainfall kinetic energy | Cumulative soil loss (g m$^{-2}$) | Percentage of soil loss (%) | Average and SD of soil loss per event (g m$^{-2}$) |
|---|---|---|---|---|
| CON | Class 1 (51) | 763.0 | 29 | 44.9 ± 42.9 b A |
| | Class 2 (66) | 786.5 | 30 | 35.7 ± 33.8 b A |
| | Class 3 (33) | 1034.0 | 40 | 94.0 ± 81.6 c B |
| HOR | Class 1 (51) | 355.4 | 34 | 20.9 ± 28.0 ab A |
| | Class 2 (66) | 364.5 | 34 | 16.6 ± 20.7 ab A |
| | Class 3 (33) | 336.3 | 32 | 30.6 ± 37.5 ab A |
| LEG | Class 1 (51) | 370.7 | 24 | 21.8 ± 16.4 b A |
| | Class 2 (66) | 534.2 | 34 | 24.3 ± 21.1 b A |
| | Class 3 (33) | 649.9 | 42 | 59.1 ± 52.4 bc B |
| BRA | Class 1 (51) | 108.3 | 23 | 6.4 ± 9.3 a A |
| | Class 2 (66) | 219.8 | 47 | 10.0 ± 15.9 a A |
| | Class 3 (33) | 143.3 | 30 | 13.0 ± 13.2 a A |

Different lowercase letters in each column mean differences between treatments within the same rainfall total energy. Different uppercase letters in each column mean differences between rainfall total energy within each treatment at $p < 0.05$. Number of cases in brackets

**Table 6** Average and standard deviation (SD) of soil loss per event, with the number of cases in brackets (*N*)

| Soil covered | Rainfall total energy | | |
|---|---|---|---|
| | Class 1 | Class 2 | Class 3 |
| <20 % | 33.9 ± 37.8 (41) a B | 28.6 ± 29.1 (42) a B | 76.7 ± 69.5 (23) b B |
| 20–40 % | 13.1 ± 14.4 (7) a AB | 18.4 ± 17.9 (23) a AB | 22.4 ± 23.6 (12) a A |
| >40 % | 7.8 ± 9.6 (20) a A | 12.2 ± 21.0 (23) a A | 14.5 ± 16.3 (9) a A |

Different lowercase letters in each row mean significant differences between rainfall total energy for the same cover. Different uppercase letters in each column mean significant differences between covers at $p < 0.05$. Soil cover: low (0–20 %), medium (20–40 %) and moderate-high (>40 %); rainfall kinetic energy (class 1: KE ≤ 2 MJ ha$^{-1}$, class 2: 2 < KE ≤ 4 MJ ha$^{-1}$ or class 3: KE > 4 MJ ha$^{-1}$)

limit ranging from 0.3 to 1.4 t ha$^{-1}$ year$^{-1}$. In shallow soils, Bienes et al. (2002) established a limit of 5 t ha$^{-1}$ year$^{-1}$, Kassam et al. (1991) reported 3 t ha$^{-1}$ year$^{-1}$ and Arnoldus (1977) published 2.2 t ha$^{-1}$ year$^{-1}$ for practically non-renewable and shallow soils. Therefore, we can say that tillage as applied in this study is not a sustainable practice, even for small scale measures. Local farmers usually till their olive groves three times per year, so the soil loss will be greater than the rate measured in this study. A permanent cover crop is clearly sustainable under current conditions; annual cover crops were at the limit.

Gómez et al. (2014) found that the greatest soil loss in an olive grove in SW Spain was related to rill and gully processes (16.1 t ha$^{-1}$ year$^{-1}$ in a 5-year period). Nevertheless, Bienes and Marques (2008) found that the soil loss in a catchment from rill processes during a highly erosive single event was only 1 to 14 % higher than the soil loss measured in the microplots. Under our experimental conditions, splash and interrill erosion are likely to be the primary factors, and thus, our rates of soil loss slightly underestimated the soil loss at the catchment level. Even so, these small plots underestimate the soil loss, being one of the limitations of the use of microplots in erosion studies as Boix-Fayos et al. (2006) pointed out. Our team (Bienes et al. 2000) recorded soil losses of up to 10.3 t ha$^{-1}$ year$^{-1}$ in bare soil from USLE erosion plots (20 × 4 m$^2$) under more intense rainfall events in the study area.

The rainfall events were more intense and had higher KE values during spring than summer, despite the great variation between events (Table 2). Because of these characteristics, the highest rates of soil loss occurred in the spring, whilst summer presented lower values in this period (Table 4). Conversely, Gómez et al. (2014) found autumn and winter to be the seasons with the greatest soil loss. In our study, differences in soil loss were statistically significant only for the CON, although the same trend can be observed in BRA. Interannual soil loss variations from water erosion were very important (Table 4), but the differences among repetitions were also high because of natural and measured variability as noted by Boix-Fayos et al. (2006) and Bagarello and Ferro (2004), with a high coefficient of variation (Gómez et al. 2001). For this reason, it was difficult to find significant differences for the different cropping seasons between treatments.

The behaviour of soils under annual cover crops was slightly different. The average soil loss during spring was quite similar to that of the autumn for LEG, whilst in HOR it was slightly lower (Table 4), possibly because of the faster growing of cereals. The protective effect of the annual cover crops that usually were at the maximum during spring (Fig. 2) protected the bare soil from the direct impact of raindrops and stopped runoff. Seeding was performed in the autumn and the soil remained bare for several weeks.

The high correlation between soil loss and cover with a negative relation was described by Dunjo et al. (2004) to the

**Table 7** Spearman's correlations between soil organic carbon (OC) and textural classes (sand, silt, clay and silt + clay) of sediments

| | OC | Sand (2000–20 μm) | Silt (20–2 μm) | Clay (<2 μm) | Silt + clay (<20 μm) |
|---|---|---|---|---|---|
| OC | | −0.700** | 0.750** | 0.200 | 0.700** |
| *P* | −0.024 | −0.055 | −0.297 | 0.527 | 0.055 |
| $I_{15}$ | 0.602 | −0.762** | 0.470 | 0.652** | 0.762** |
| $I_{30}$ | 0.635*** | −0.796* | 0.620** | 0.450 | 0.796* |
| *I* | 0.627*** | −0.710** | 0.440 | 0.599*** | 0.710** |
| KE | 0.190 | −0.127 | −0.212 | 0.552*** | 0.127 |
| *R* | 0.262 | −0.491 | 0.139 | 0.636** | 0.491 |

*P* rainfall amount, $I_{15}$ maximum intensity in 15 min, $I_{30}$ maximum intensity in 30 min, *I* maximum intensity in 1 h, *KE* kinetic energy, *R factor* erosivity

*$p < 0.01$; **$p < 0.05$; ***$p < 0.10$

**Table 8** Average and standard deviation (SD) of textural fractions and OC of the reference soil ($Ap_1$) and sediments of two groups according to the rainfall intensity, which was moderate $2 < I \leq 20$ mm h$^{-1}$ and high intensity $20 < I < 50$ mm h$^{-1}$

| | Reference soil ($Ap_1$) ($N = 4$) | Sediments grouped by rainfall intensity | |
|---|---|---|---|
| | | Moderate ($N = 5^a$) | High ($N = 5$) |
| $P$ (mm) | | 34 ± 12 | 49 ± 22 |
| $I_{15}$ (mm h$^{-1}$) | | 6.5 ± 1.7 a | 21.0 ± 10.3 b |
| $I_{30}$ (mm h$^{-1}$) | | 5.7 ± 1.5 a | 13.3 ± 6.5 b |
| $I$ (mm h$^{-1}$) | | 7.9 ± 2.0 a | 29.3 ± 12.3 b |
| KE (MJ ha$^{-1}$) | | 4.8 ± 1.7 a | 8.5 ± 3.5 b |
| $R$ (MJ mm ha$^{-1}$ h$^{-1}$) | | 15.0 ± 5.3 a | 68.5 ± 42.7 b |
| Sand (2000–20 μm) (g kg$^{-1}$)) | 5.8 ± 1.1 | 4.59 ± 1.4 | 3.8 ± 0.9 |
| Silt (20–2 μm) (g kg$^{-1}$) | 2.9 ± 1.4 | 3.3 ± 1.2 | 4.0 ± 1.2 |
| Clay (<2 μm) (g kg$^{-1}$) | 1.2 ± 0.2 a | 1.7 ± 0.6 ab | 2.2 ± 0.6 b |
| Silt + clay (<20 μm) (g kg$^{-1}$) | 4.1 ± 1.2 | 5.0 ± 1.4 | 6.2 ± 0.9 |
| OC (%) | 0.6 ± 0.1 a | 1.3 ± 0.4 ab | 1.6 ± 0.4 b |

The number of samples is given in the brackets. Different lowercase letters in rows mean significant differences between groups at $p < 0.05$

$P$ rainfall amount, $I_{15}$ maximum intensity in 15 min, $I_{30}$ maximum intensity in 30 min, $I$ maximum intensity in 1 h, *KE* kinetic energy, *R factor* erosivity

[a] $N = 5$, except for OC that was $N = 3$

same degree. In the spring, both annual and permanent cover crops covered the soil in different proportions but always over CON (Fig. 2), but in all cases, it was enough to protect the soil from rainfall erosivity. This is the reason why it is vital to inform the farmers that maintaining minimum cover during the spring and autumn will limit soil and fertility loss (Gómez et al. 2009b). Thus, for gypsiferous soils in a semiarid climate, it should be compulsory to protect the soil against erosion in the spring and autumn; such prevention measures must be adapted to the particular climate (Fleskens and Stroosnijder 2007).

Although the high-energy events (class 3: KE > 4 MJ ha$^{-1}$) only made up 22 % of the total, they drove to the highest proportion of soil loss in CON and LEG, at 40 and 42 %, respectively (Table 5), primarily because of the high proportion of bare soil during these events. The worst situation was the loss of soil under CON treatment from high-energy events (Table 5), in other words, with low soil cover (Table 6) similar to the results of Martínez-Mena et al. (2012). Soil loss under class 2 (2 < KE ≤ 4 MJ ha$^{-1}$) events was significantly reduced with at least 40 % soil cover, supporting the results of Marques et al. (2008) at cover percentages above 60 %. For south of Spain, Espejo-Perez et al. (2013) determined that with a cover of over 20 % soil was well protected against more events.

The important effect of protecting soil against erosion is crucial under low cover percentages. In semiarid gypsiferous soils, it is very difficult to increase the soil cover beyond 55 %. Nevertheless, in this study, we note the necessity of soil cover, even at low values, and how much soil erosion can be reduced with these cover crops (Fig. 4). Furthermore, the effort and economic investments needed to achieve higher covers seem to justify the soil loss reduction, although water and nutrient balance were not studied in this work. Nevertheless, similar research of permanent and mowed cover crops in vineyards near the study region (Ruiz-Colmenero et al. 2011, 2013) demonstrated that the loss of SOC, nitrogen and phosphorus was reduced by 50 %; runoff was also reduced between 75 and 84 %. However, soil moisture was not higher due to the water used by the crops.

In this study, we did not record any event over 50 mm h$^{-1}$, although these events are not considered infrequent (García-Díaz et al. 2015). In 2007, our team recorded a single event in the study area ($P = 43$ mm, $I = 55$ mm h$^{-1}$, return period of 7 years according to Elías-Castillo and Ruiz Beltrán (1979)), which produced a soil loss of 93.47 t ha$^{-1}$ in a sloping olive grove managed by minimum tillage (Bienes et al. 2012). In Central Spain, García-Díaz et al. (2015) found that there was a progressive increase in the $I$ and $R$ factors from 1995 until 2012, whilst the annual rainfall amount remained similar. As Nearing et al. (2005) found, the higher the intensity of the rainfall event, the more soil was lost by erosion (Table 5). Unless the situation changes, the problem of erosion in the area of study will be worse in the future.

Given that erosion is a selective process (Martinez-Mena et al. 2001), changes in texture are frequently mentioned in the literature. Martinez-Mena et al. (2002) found less sand (>50 μm) in sediments that were eroded after rain events of high or medium intensity than low-intensity events, which was similar to our results (Table 8). Different authors find different particle size enrichment, e.g. Martinez-Mena et al. (2002) and Pieri et al. (2009) describe silt enrichment (2–20 μm). In this study, the primary change was found for clay fractions (Table 8) with an enrichment rate of 1.6 on average.

**Table 9** Farmer and exploitation profiles and a brief review of the biophysical factors affecting the soil as identified by land users. The justifications for their management practices and some socioeconomic elements affecting the adoption of SLM practices

| | Results of the survey |
|---|---|
| No. of farmers | 119 |
| Age | 54 ± 14 years old (mean ± SD) |
| Gender | 86 % males and 14 % females |
| Level of studies | Primary 42.0 %<br>Secondary 22.7 %<br>Professional training 16.8 %<br>University 18.5 % |
| Reasons for being involved in farming activities | Family tradition (81 %)<br>Vocation (13 %)<br>Others (6 %) |
| Education about soils | 77 % of farmer's declare not having received any education or training with regard to soils |
| Sources of income | 48 % of farmers have other activities to obtain their incomes, mainly in the third sector (services: transport, leisure, etc.) |
| Agrarian subsidies | 98 % receive subsidies to support farming activities. However, the number of aids related to soil conservation or ecological/organic practices is below 17 % |
| Farm size number of ha | Mean 184 ha, SD 328 (min 1 ha; max 1700 ha) |
| No. of municipalities where they have plots | Mean 2, SD 1.6 (min 1; max 12) |
| Crops | The most common crops are olive groves, rain-fed cereals, irrigated cereals, vineyards and horticultural crops, in decreasing order. Farmers tend to grow different crops. The most common combination is olive groves + rain-fed cereals + vineyards |
| No. of olive growers | 83 |
| Reasons for their management practices | Tradition 50 %<br>Profitability 31 %<br>Product quality 24 %<br>Environmental sustainability 18 %<br>Known practices 18 %<br>Typical in the region 17 %<br>Subsidies 7 %<br>Commercial brands advice 2 % |
| Bio-physical factors affecting soil degradation identified by land users | In order of importance farmers are able to acknowledge: steep slopes, waterlogging, erosion, low water retention, low fertility, compaction, soil organic matter depletion, pollution, salinisation, alkalinity and acidity |
| Soil improvement measures | 79 % of farmers consider that they can contribute to solve soil problems especially by:<br>Increasing soil OC 40 %<br>Tillage improvement 21 %<br>Improving the use of fertilisers 13 %<br>Land levelling works 12 %<br>More sustainable land management practices 9 %<br>Improving the use of pesticides 2 % |
| Use of cover crops in olive groves | 5/83 |
| What do they need to improve soil management | Training 29 %<br>Better technology 21 %<br>Institutional aids 19 % |
| Training demands | 62 % of farmers would like to participate in trainings related to:<br>Oliviculture 16 %<br>Rain-fed cereals 11 %<br>Viticulture 7 %<br>Organic farming 7 %<br>Pesticides 6 %<br>Tillage 3 %<br>Pruning 2 %<br>Institutional aids 3 %<br>Horticulture 2 % |
| Sources of agrarian information | Neighbours and acquaintances (27 %)<br>Regional extension services (26 %)<br>Internet (14 %)<br>Commercial agricultural suppliers (12 %)<br>Farmers associations (10 %)<br>Specialised publications, newspapers or university professionals (<4 %) |

Although the sediment fractions and rainfall intensity ($I$, $I_{15}$ and $I_{30}$) were correlated (Table 7), no significant differences appeared among the high and medium events. Despite that finding, there was an increasing trend in fine-sized sediment particles with high-intensity events. This finding can be explained by the aggregate breakdown under high-intensity events in which the splash is the primary erosion process. Martinez-Mena et al. (2002) found a decrease in coarse fractions with runoff time for high-intensity events (56.0 ± 2.4 mm $h^{-1}$), whereas a time-independent size distribution of the eroded material indicated transport-limited erosion under medium-intensity events (31.4 ± 1.4 mm $h^{-1}$). In addition, Wang et al. (2014) found that a higher KE (>105 J $m^{-2}$ $h^{-1}$) resulted in a higher abundance of fine particles, supporting our results (Table 8).

The OC mass of the sediments was 1.5 % on average, a little less than the 1.8 % described by Marques et al. (2008) and higher than the 1.21 % found by Gómez et al. (2009b). This 1.5 % implies an enrichment rate of 2.4, which is similar to the 2.38 reported by Martinez-Mena et al. (2008) in an olive grove in SE Spain. There are other studies that report a similar OC enrichment of sediments (e.g. Martínez-Mena et al. 2012; Marquez-Garcia et al. 2013). This enrichment of the sediments in clay fractions and OC is from organo-mineral associations (Porta et al. 1999). Rainfall events of higher intensity

($I_{30}$ or $I$) led to particularly greater OC loss (Table 8), according to the results of Martinez-Mena et al. (2008). Although statistically significant differences did not appear when sediments from events of different intensities were compared, there seems to be a trend. These results were the opposite of those of Jacinthe et al. (2004), who found more OC in the sediments that were mobilised under low-intensity storms in different watersheds. In any case, when considering the basin scale, this process leads to a decline in soil fertility on-site, and water pollution off-site, from the rich OC-suspended sediments that were transported by runoff (Pieri et al. 2009).

There are other benefits of no tillage managements to improve the sustainability of olive groves. In addition to soil loss control, there is an increase in soil organic matter and therefore carbon sequestration (Parras-Alcantara et al. 2015); an increase in water and nutrient availability (Colen et al. 2016); and a promotion and conservation of soil quality and health, as well the stability of ecosystems (Correia et al. 2015).

Despite the environmental benefits reported by scientific evidence, agricultural soil degradation is not obvious to farmers, and therefore, changing their management practices is not a priority for them. This general statement has been acknowledged in the region where less than one third of farmers would be willing to change their traditional tillage practice by cover crops (Marques et al. 2015). Currently, only 6 % of farmers who participated in the survey were using cover crops (Table 9). Moreover, according to the results, there is little probability of spontaneous cover crop adoption in this region.

The poor understanding of long-term land degradation processes by farmers, the lack of a holistic perspective to propose flexible and suitable solutions adapted to the local contexts as well as insufficient use of land users' own experiences have been identified as major constraints for the adoption of SLM practices (Liniger et al. 2004). Remarkably, the farmers acknowledge their need for training and highlight the importance of extension services for obtaining information; therefore, there is a need to strengthen cooperation among farmers, researchers and agrarian extension services to facilitate decision support. Conversely in Ethiopia, Assefa and Hans-Rudolf (2015) found that farmers have a wealth of experience in identifying the severity of soil erosion and its effect on soil fertility decline, even so farmers abandoned the tradition of fallowing. Vila et al. (2015) identified a lack of consensus on land management priorities, being necessary to promote participative methods and incorporate environmental service assessment as key elements in the decision-making processes. A similar conclusion found Teshome et al. (2014) highlighting that policy makers should take into consideration land-related factors in designing and implementing SLM policies and programmes to influence farmers' investments in SLM practices.

The results of this survey prioritise the need for information over access to governmental aid or subsidies. This finding supports recent literature that questions the efficiency of subsidies because there is no simple relation between financial reward and behavioural change (de Snoo et al. 2013; Merckx and Pereira 2015).

## 5 Conclusions

Olive groves that are managed under tillage produced unsustainable rates of soil loss, even at small scales, under the study conditions. Gypsiferous soils are vulnerable to water erosion processes if they are not protected by vegetation, even under moderate intensity rainfalls. This study shows that soil loss is correlated to scarcity of vegetation cover and kinetic energy of rainfall. Spring and autumn had higher intensity rainfalls, and therefore, soil loss is also higher during these seasons, primarily in the spring. Farmers should adopt tailored soil cover practices during those periods.

Permanent cover crops (*B. distachyon*) in olive groves demonstrated their efficiency to prevent water erosion. After 1 year, the establishment of this cover is enough to reach a sediment yield that can be considered tolerable. Annual cover crops (*H. vulgare* and *V. ervilia*) were also efficient during the spring, but they needed time to be established unlike *O. viciifolia*, which was not suitable for this area because it did not grow properly during the dry seasons.

The OC in the soil was lost with fine soil fractions, enriching sediments in OC (average concentration 1.5 %). This finding underscores the current state of organic matter depletion characteristics in these poor and shallow soils. Erosion not only implies soil loss particles but a decrease in OC which results in the loss of agricultural land due to abandonment and reduced productivity.

Farmers must be conscious of the importance of vegetation cover. Cover crops in olive groves are rare in this region because farmers are reluctant to change from traditional tillage to cover crops due to the risk of competition for water and nutrient resources. Their practices are driven by tradition and economics. Change to cover crops would be possible if there was an economic profit and not only an environmental benefit. In this region, subsidies are given limited importance in achieving changes in farmer behaviour with regards to management practice decisions. Farmers easily understand and declare the need for active practices to conserve soil organic matter; this recognition can be used as a link to introduce some other important needs to improve physical chemical soil quality. In this context, the work of extension agents and institutions responsible for education and training is considered important, particularly to increase awareness of the use of cover crops and their influence on olive production and the environment.

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## References

Alcántara C, Pujadas A, Saavedra M, Alcántara C (2011) Management of cruciferous cover crops by mowing for soil and water conservation in southern Spain. Agric Water Manag 98:1071–1080

Arnoldus H (1977) Methodology used to determine the maximum potential average annual soil loss due to sheet and rill erosion in Morocco. FAO Soils Bulletins (FAO)

Assefa E, Hans-Rudolf B (2015) Farmers' perception of land degradation and traditional knowledge in Southern Ethiopia—resilience and stability. Land Degrad.Dev 27:1552–1561

Bagarello V, Ferro V (2004) Plot-scale measurement of soil erosion at the experimental area of Sparacia (southern Italy). Hydrol Process 18: 141–157

Bienes R, Guerrero-Campo J, Aroca JA, Gómez B, Nicolau JM, Espigares T (2000) Runoff coefficient and soil erosion rates in croplands in a Mediterranean-continental region in Central Spain. In: Rubio JL, Morgan RPC, Andreu V (eds) Proceedings of the Third International Congress Man and Soil at the Third Millennium, 28 March- 1 April 2000. Geoforma, Logroño, pp 1359–1368

Bienes R, Pérez R, Gómez B (2002) La erosión. Agricultura

Bienes R, Marques MJ (2008) Rill and interril erosion produced by a single-storm event in an olive grove in Central Spain. Proceedings of EUROSOIL 2008, Vienna, Austria, 25–29 August 2008

Bienes R, Ruiz M, Marques MJ (2010) Pérdida de suelo, fósforo y materia orgánica por erosión hídrica en parcelas revegetadas con matorral autóctono bajo clima semiárido [Loss of soil, phosphorous and organic matter by hydric erosion in revegetated plots with authoctonous shrubs under semiarid climate]. Revista de Ciências Agrárias 33:58–69

Bienes R, Marques MJ, Ruiz-Colmenero M (2012) Cultivos herbáceos, Viñedos y Olivares. El manejo tradicional del suelo y sus consecuencias en la erosión hídrica [Grass crops, vineyards and olive groves. The traditional management of soil and its consequences on hydric erosion]. Cuadernos de Investigación Geográfica 38:49–74

Boix-Fayos C, Arnau-Rosalén E, Calvo-Cases A, Castillo V, Albaladejo J (2006) Measuring soil erosion by field plots: understanding the sources of variation. Earth-Sci Rev 78:267–285

Calatrava-Leyva J, Franco JA, Gonzalez-Roa MC (2007) Analysis of the adoption of soil conservation practices in olive groves: the case of mountainous areas in southern Spain. Span J Agric Res 5:249–258

Celette F, Gaudin R, Gary C (2008) Spatial and temporal changes to the water regime of a Mediterranean vineyard due to the adoption of cover cropping. Eur J Agron 29:153–162

Colen L, Turkelboom F, Van Steenwinkel S, Al Ahmed K, Deckers J, Poesen JP (2016) How the soil moves upward in the olive orchards of NW Syria: sustainability analysis of a local innovation. Land DegradDev 27:416–426

Correia CM, Brito C, Sampaio A, Dias AA, Bacelar E, Gonçalves B, Ferreira H, Moutinho-Pereira J, Rodrigues MA (2015) Leguminous cover crops improve the profitability and the sustainability of rainfed olive (*Olea europaea* L.) orchards: from soil biology to physiology of yield determination. Procedia Environ Sci 29: 282–283

de Snoo GR, Herzon I, Staats H, Burton RJF, Schindler S, van Dijk J, Lokhorst AM, Bullock JM, Lobley M, Wrbka T, Schwarz G, Musters CJM (2013) Toward effective nature conservation on farmland: making farmers matter. Conserv Lett 6:66–72

Dunjo G, Pardini G, Gispert M (2004) The role of land use-land cover on runoff generation and sediment yield at a microplot scale, in a small Mediterranean catchment. J Arid Environ 57:239–256

Duran-Zuazo VH, Rodriguez-Pleguezuelo CR (2008) Soil-erosion and runoff prevention by plant covers. A review. Agron Sustain Dev 28:65–86

EEA (2015) The European environment—state and outlook 2015: synthesis report. Copenhagen. doi:10.2800/944899

Elías-Castillo F, Gimenez Ortiz R (1965) Evapotranspiraciones potenciales y balances de agua en españa. [Potential evapotranspiration and water balance in Spain]. Direccion General de Agricultura, Ministerio de Agricultura, Madrid

Elías-Castillo F, Ruiz Beltrán L (1979) Precipitaciones máximas de España. Madrid

Espejo-Perez AJ, Rodriguez-Lizana A, Ordonez R, Giraldez JV (2013) Soil loss and runoff reduction in olive-tree dry-farming with cover crops. Soil Sci Soc Am J 77:2140–2148

FAOSTAT (2015) Food and Agriculture Organization of the United Nations-FAOSTAT database. In: http://faostat3.fao.org. Accessed 12 Jan 2015

Faugier J, Sargeant M (1997) Sampling hard to reach populations. J Adv Nurs 26:790–797

Fleskens L, Stroosnijder L (2007) Is soil erosion in olive groves as bad as often claimed? Geoderma 141:260–271

Franco JA, Calatrava-Leyva J (2006) Adoption of soil erosion control practices in southern Spain olive groves. Proceedings of the International Association of Agricultural Economists, Gold Coast, Australia, 12–18 August 2006

García-Díaz A, Bienes R, Sastre B (2015) Study of climatic variations and its influence on erosive processes in recent decades in one location of Central Spain. In: Lolllino et al (eds) Engineering geology for society and territory—volume 1. Springer International Publishing, Switzerland, pp. 105–108

García-Estríngana P, Alegre Álvaro J, Alonso Blázquez N, Guerrero E (2005) Soil coverage measure methods in small plots assays [Determinación de la cobertura del suelo en ensayos de pequeña parcela una comparación de métodos]. In: Jimenez Ballesta R, Alvarez Gonzalez AM (eds) Control de la degradación de suelos. Madrid, pp 275–279

Gerlach T (1967) Hillslope for measuring sediment movement. Revue Geomorphology Dynamic 4:173–175

Gessesse B, Bewket W, Bräuning A (2015) Model-based characterization and monitoring of runoff and soil erosion in response to land use/land cover changes in the Modjo Watershed. Ethiopia Land DegradDev 26:711–724

Gómez JA, Nearing M, Giráldez JV, Alberts E (2001) Analysis of sources of variability of runoff volume in a 40 plot experiment using a numerical model. J Hydrol 248:183–197

Gómez JA, Guzmán MG, Giráldez JV, Fereres E (2009a) The influence of cover crops and tillage on water and sediment yield, and on nutrient, and organic matter losses in an olive orchard on a sandy loam soil. Soil Till Res 106:137–144

Gómez JA, Sobrinho TA, Giráldez JV, Fereres E (2009b) Soil management effects on runoff, erosion and soil properties in an olive grove of Southern Spain. Soil Till Res 102:5–13

Gómez JA, Llewellyn C, Basch G, Sutton PB, Dyson JS, Jones CA (2011) The effects of cover crops and conventional tillage on soil and runoff loss in vineyards and olive groves in several Mediterranean countries. Soil Use Manage 27:502–514

Gómez JA, Vanwalleghem T, De Hoces A, Taguas EV (2014) Hydrological and erosive response of a small catchment under olive cultivation in a vertic soil during a five-year period: implications for sustainability. Agric Ecosyst Environ 188:229–244

IECM (2013) Instituto de Estadística de la Comunidad de Madrid-DESVAN. Banco de Datos Estructurales. http://www.madrid.

org/desvan/desvan/AccionDatosUnaSerie.icm?codSerie=27115 Accessed 18 Aug 2015

INE (2002) National Statistics Institute—Instituto Nacional de estadística. In: http://www.ine.es. Accessed 15 Mar 2015

IUSS Working Group WRB (2014) World Reference Base for Soil Resources 2014. International soil classification system for naming soils and creating legends for soil maps. In: World Soil Resources Reports n° 106. FAO, Rome

Jacinthe P, Lal R, Owens LB, Hothem DL (2004) Transport of labile carbon in runoff as affected by land use and rainfall characteristics. Soil Tillage Res 77:111–123

Kairis O, Karavitis C, Kounalaki A, Salvati L, Kosmas C (2013) The effect of land management practices on soil erosion and land desertification in an olive grove. Soil Use Manag 29(4):597–606. doi:10.1111/sum.12074

Karamesouti M, Detsis V, Kounalaki A, Vasiliou P, Salvati L, Kosmas C (2015) Land-use and land degradation processes affecting soil resources: evidence from a traditional Mediterranean cropland (Greece). Catena 132:45–55

Kassam AH, Fischer G, Antoine J (1991) Agro-ecological land resources assessment for Agricultural Development Planning: a case study of Kenya. Resources data base and land productivity. Technical Annex 2. Soil erosion and productivity. FAO/IIASA, Rome, ISBN 92–5103303-X

Keesstra S, Pereira P, Novara A, Brevik EC, Azorin-Molina C, Parras-Alcantara L, Jordan A, Cerda A (2016) Effects of soil management techniques on soil water erosion in apricot orchards. Sci Total Environ 551:357–366

Ligonja PJ, Shrestha RP (2015) Soil erosion assessment in Kondoa eroded area in Tanzania using universal soil loss equation, geographic information systems and socioeconomic approach. Land Degrad Dev 26:367–379

Liniger H, Douglas M, Schwilch G (2004). Towards sustainable land management—'common sense' and some of the other key missing elements (the WOCAT experience). Proceedings of the 13th International Soil Organization Conference (ISCO). Conserving Soil and Water for Society: Sharing Solutions. 1–6, Brisbane, Australia, 4–8 July 2004

MAGRAMA (2013) Encuesta de Superficies y Rendimientos de Cultivos (ESYRCE). Análisis de las plantaciones de olivar en España [Survey about extensions and yields in crops (ESYRCE). Analysis of olive groves in Spain]. Secretaría General Técnica, Subdirección General de Estadística del Ministerio de Agricultura, Alimentación y Medio Ambiente, Madrid

MAPA (1994) Métodos Oficiales de Análisis, Tomo III [Official methods of analysis. Book III], Madrid

Marques MJ, Bienes R, Perez-Rodriguez R, Jimenez L (2008) Soil degradation in Central Spain due to sheet water erosion by low-intensity rainfall events. Earth Surf Process Landforms 33:414–423

Marques MJ, Bienes R, Cuadrado J, Ruiz-Colmenero M, Barbero-Sierra C, Velasco A (2015) Analysing perceptions attitudes and responses of winegrowers about sustainable land management in Central Spain. Land Degrad Dev 26:458–467

Marquez-Garcia F, Gonzalez-Sanchez EJ, Castro-Garcia S, Ordonez-Fernandez R (2013) Improvement of soil carbon sink by cover crops in olive orchards under semiarid conditions. Influence of the type of soil and weed. Span J Agric Res 11:335–346

Martinez-Mena M, Castillo V, Albaladejo J (2001) Hydrological and erosional response to natural rainfall in a semi-arid area of south-east Spain. Hydrol Process 15:557–571

Martinez-Mena M, Castillo V, Albaladejo J (2002) Relations between interrill erosion processes and sediment particle size distribution in a semiarid Mediterranean area of SE of Spain. Geomorphology 45: 261–275

Martinez-Mena M, Lopez J, Almagro M, Boix-Fayos C, Albaladejo J (2008) Effect of water erosion and cultivation on the soil carbon stock in a semiarid area of south-east Spain. Soil Till Res 99:119–129

Martinez-Mena M, López J, Almagro M, Albaladejo J, Castillo V, Ortiz R, Boix-Fayos C (2012) Organic carbon enrichment in sediments: effects of rainfall characteristics under different land uses in a Mediterranean area. Catena 94:36–42

Merckx T, Pereira HM (2015) Reshaping agri-environmental subsidies: from marginal farming to large-scale rewilding. Basic Appl Ecol 16: 95–103

Nearing MA, Jetten V, Baffaut C, Cerdan O, Couturier A, Hernandez M, Le Bissonnais Y, Nichols MH, Nunes JP, Renschler CS, Souchere V, van Oost K (2005) Modeling response of soil erosion and runoff to changes in precipitation and cover. Catena 61:131–154

Ochoa PA, Fries A, Mejia D, Burneo JI, Ruiz-Sinoga JD, Cerda A (2016) Effects of climate, land cover and topography on soil erosion risk in a semiarid basin of the Andes. Catena 140:31–42

Ordóñez-Fernández R, Rodriguez-Lizana A, Espejo-Pérez AJ, González-Fernández P, Saavedra MM (2007) Soil and available phosphorus losses in ecological olive groves. Eur J Agron 27:144–153

Palese AM, Ringersma J, Baartman JEM, Peters P, Xiloyannis C (2015) Runoff and sediment yield of tilled and spontaneous grass-covered olive groves grown on sloping land. Soil Res 53:542–552

Panagos P, Meusburger K, Ballabio C, Borrelli P, Alewell C (2014) Soil erodibility in Europe: a high-resolution dataset based on LUCAS. Sci Total Environ 479–480:189–200

Panagos P, Ballabio C, Borrelli P, Meusburger K, Klik A, Rousseva S, Tadić MP, Michaelides S, Hrabalíková M, Olsen P, Aalto J, Lakatos M, Rymszewicz A, Dumitrescu A, Beguería S, Alewell C (2015) Rainfall erosivity in Europe. Sci Total Environ 511:801–814

Parras-Alcantara L, Diaz-Jaimes L, Lozano-Garcia B (2015) Organic farming affects C and N in soils under olive groves in Mediterranean areas. Land Degrad Dev 26:800–806

Pastor M (2008) Sistemas de manejo del suelo [Soil management systems]. In: Barranco D, Fernández-Escobar R, Rallo L (eds) El cultivo del olivo, 6ª edn. Junta de Andalucía, Consejería de Agricultura y Pesca & Ediciones. Mundi-Prensa, Madrid, pp. 239–295

Pieri L, Bittelli M, Hanuskova M, Ventura F, Vicari A, Pisa PR (2009) Characteristics of eroded sediments from soil under wheat and maize in the North Italian Apennines. Geoderma 154:20–29

Porta J, López-Acevedo M, Roquero C (1999) Edafología para la agricultura y el medio ambiente [Edaphology for agriculture and environment]. Ediciones Mundi-Prensa, Madrid

Prosdocimi M, Cerda A, Tarolli P (2016) Soil water erosion on Mediterranean vineyards: a review. Catena 141:1–21

Rallo L, Barranco D, Caballero JM, del Río C, Martín A, Tous J, Trujillo I (2005) Variedades de olivo en España [Olive tree cultivars in Spain]. Junta de Andalucía, MAPA y Ediciones. Mundi-Prensa, Madrid

Rodrigo-Comino J, Iserloh T, Morvan X, Malam Issa O, Naisse C, Keesstra SD, Cerdà A, Prosdocimi M, Arnáez J, Lasanta T, Ramos MC, Marques MJ, Ruiz Colmenero M, Bienes R, Ruiz Sinoga JD, Seeger M, Ries JB (2016) Soil erosion processes in European vineyards: a qualitative comparison of rainfall simulation measurements in Germany. Spain and France Hidrology 3. doi:10.3390/hydrology3010006

Ruiz-Colmenero M, Bienes R, Marques MJ (2011) Soil and water conservation dilemmas associated with the use of green cover in steep vineyards. Soil Till Res 117:211–223

Ruiz-Colmenero M, Bienes R, Eldridge DJ, Marques MJ (2013) Vegetation cover reduces erosion and enhances soil organic carbon in a vineyard in the Central Spain. Catena 104:153–160

Taguas EV, Ayuso JL, Pérez R, Giráldez JV, Gómez JA (2013) Intra and inter-annual variability of runoff and sediment yield of an olive micro-catchment with soil protection by natural ground cover in Southern Spain. Geoderma 206:49–62

Taguas EV, Gómez JA (2015) Vulnerability of olive orchards under the current CAP (Common Agricultural Policy) regulations on soil erosion: a study case in Southern Spain. Land Use Policy 42:683–694

Teshome A, de Graaff J, Ritsema C, Kassie M (2014) Farmers' perceptions about the influence of land quality, land fragmentation and tenure systems on sustainable land management in the north western Ethiopian highlands. Land Degrad Dev 27:884–898

Urbano P (1992) Tratado de fitotecnia general [Treaty on general plant breeding]. Mundi-Prensa, Madrid

Verheijen FGA, Jones RJA, Rickson RJ, Smith CJ (2009) Tolerable versus actual soil erosion rates in Europe. Earth-Sci Rev 94:23–38

Vila J, Rodríguez-Carreras R, Varga D, Ribas A, Úbeda X, Asperó F, Llausàs A, Outeiro L (2015) Stakeholder perceptions of landscape changes in the Mediterranean mountains of the North-Eastern Iberian Peninsula. Land Degrad Dev 27:1354–1365

Walkley A, Black IA (1934) An examination of Degtjareff method for determining soil organic matter and a proposed modification of the chromic acid titration method. Soil Sci 37: 29–38

Wang L, Shi ZH, Wang J, Fang NF, Wu GL, Zhang HY (2014) Rainfall kinetic energy controlling erosion processes and sediment sorting on steep hillslopes: a case study of clay loam soil from the Loess Plateau, China. J Hydrol 512:168–176

Wischmeier WH, Smith DD (1978) Predicting rainfall erosion losses. A guide to conservation planning. Agriculture handbook no. 537. USDA, Washington DC

Zema DA, Denisi P, Taguas Ruiz EV, Alfonso Gomez J, Bombino G, Fortugno D (2016) Evaluation of surface runoff prediction by AnnAGNPS model in a large Mediterranean watershed covered by olive groves. Land Degrad Dev 27:811–822